

PRANJAL SINGH

Full Stack Software Developer (MERN) | Gen AI & LLM Integration | AI Agents & Automation | RAG

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PROFESSIONAL SUMMARY

Results-driven Full Stack Software Developer and B.Tech CSE student at AKTU with hands-on experience building and deploying full-stack web applications and AI agent automation systems using React.js, Node.js, Express.js, MongoDB, Docker, and n8n. Proficient in Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, AI Agent Orchestration, and Vector Databases. Experienced with REST API design, JWT/OAuth2 authentication, containerized deployment (Docker), CI/CD, and cloud hosting via Vercel and Render. Seeking a Software Developer / Full Stack Developer internship to contribute to real-world, production-grade products.

TECHNICAL SKILLS

Programming Languages: JavaScript (ES6+), Java

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS, Responsive Design, API Integration

Backend: Node.js, Express.js, REST API Development, JWT Authentication & Authorization, OAuth2, Role-Based Access Control (RBAC), OTP Verification, SQL, MongoDB, Mongoose

Gen AI & ML: Large Language Models (LLMs), Prompt Engineering, RAG (Retrieval-Augmented Generation), Vector Databases, Embeddings, AI Agents, Gemini API (Gemini 2.5 Flash, Embedding 001), Groq API, OpenAI API, Speech Recognition API

Automation & Integration: n8n Workflow Orchestration, Webhooks, Gmail API, Google Calendar API, Slack API, GitHub API, Jira API, Notion API

Databases: MongoDB Atlas, MongoDB Atlas Vector Search, SQL, NoSQL Database Design

DevOps & Tools: Docker, Git, GitHub, CI/CD (Vercel/Render pipelines), Postman, Vercel, Render, PDF Processing, Text Extraction

PROJECTS

AI Employee Digital Twin | *In Progress* | *Expected Completion: Aug 2026*

Tech Stack: React.js, Node.js, Express.js, Docker, n8n, OpenAI API, JWT, OAuth2, Gmail/Slack/GitHub/Jira/Calendar/Notion APIs

- Designing an AI agent-based workforce automation platform that performs employee-like daily tasks such as email triage, calendar checks, Slack/Teams summarization, Jira updates, GitHub PR analysis, and automated daily reporting across 6+ integrated services.
- Completed UI/UX design for the platform; developing secure authentication (Email/Password, OTP, Google OAuth2, JWT, RBAC) and a Dockerized backend for consistent local and production environments.
- Architecting n8n-based workflows to connect Gmail, Calendar, Slack, GitHub, Jira, and Notion into a unified AI agent layer for autonomous task execution.

AI Mock Interview Platform

Tech Stack: React.js, Node.js, Express.js, MongoDB Atlas, Gemini API, JWT, RAG, Vector Search, Docker

- Built a full-stack AI-powered mock interview platform simulating real interview experiences through dynamic, role-specific question generation using the Gemini API and Prompt Engineering.
- Implemented a RAG pipeline with Gemini Embedding 001 and MongoDB Atlas Vector Search, improving relevance of retrieved context for AI-generated follow-up questions.
- Built secure JWT-based authentication with interview history tracking and personalized dashboards; containerized with Docker and deployed via Vercel and Render with CI/CD, cutting manual deployment steps.

Virtual Assistant

Tech Stack: React.js, Node.js, Express.js, MongoDB, Groq API, Bcrypt

- Developed a JARVIS-style AI virtual assistant using the MERN stack with real-time voice input/output via the Web Speech API and low-latency responses powered by the Groq API.
- Implemented secure authentication using JWT and Bcrypt with hashed credential storage.
- Added personalization features including custom assistant name and profile image upload, improving user engagement.

EDUCATION

B.Tech in Computer Science & Engineering (CSE) — AKTU | Pursuing

Senior Secondary Education (Class XII) — CBSE Board | 2023

Secondary Education (Class X) — CBSE Board | 2021

CERTIFICATIONS & ACHIEVEMENTS

- AI Workshop — COMET, IIT Roorkee (2025): Gained hands-on exposure to applied AI and machine learning techniques.
- GEN-AI Camp — AlgoUniversity (2026): Covered Generative AI, Multimodal AI Systems, and NLP.
- Java Programming Course — NPTEL (2025): Completed certified course in Java programming and object-oriented design.
- Full Stack MERN + Gen AI Projects: Built and deployed 3 production-ready applications combining the MERN stack with generative AI capabilities.
- Cloud & Container Deployment: Implemented Docker-based containerization and CI/CD workflows using Vercel and Render.